

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11.	(a)			moles water = $1.44 \div 18.02 = 0.0799$ (1) d = 4 (1) [allow ecf]		1	1	2	2	2
	(b)	(i)		bromide is present (1) the sample produces an orange-brown solution which could be bromine or iodine but cannot contain iodide as no H ₂ S produced (1)			1 1	2		2
		(ii)		moles Br = 0.0203 (1) [allow moles I = 0.01623 as ecf] moles compound = 5.076×10^{-3} so c = 4 (1) [allow 3.2 as ecf for iodide]		1	1	2	2	2
	(c)			aluminium is amphoteric	1			1		1
	(d)			$6.34\% \times 425.62 = 27.0$ shows b = 1 (1) $1.63\% \times 425.62 = 6.94$ so must be Li and only one present (1) therefore formula is LiAlBr ₄ .4H ₂ O [allow ecf from earlier parts] (1)		1 1	1	3	1 1	
				Question 11 total	1	4	5	10	6	7